MAKINI CALEB
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- BSc Mathematics
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MAKINI CALEB — Live Analytics

BSc Mathematics · Data Scientist · Every chart and prediction is generated live in R.

Live Plots | New Predictions | Forecasts | Explainable AI

12
Analysis Modules

30+
Live Charts

18
Predictors

100%
Runnable

🎯 Every model has a live prediction form

- 🤖 **KNN, Logistic, SVM** — enter values, get classification
- 🌲 **Random Forest** — regression (MPG) & classification (Iris)
- ⚡ **XGBoost** — diabetes prediction with probability
- 💡 **XAI** — global + local explanation
- ⏳ **Time Series** — interactive ARIMA forecast
- 🛒 **Project** — full e-commerce analytics

🧹 Data Cleaning — Before & After

NA by Column

Numeric Distributions (After)

Outliers Detected (IQR)

Cleaned Data Preview

🔎 Wrangling, Joins & Reshaping

Filter — MARRIED / DIVORCED

Six Join Types

- Inner
- Left
- Right
- Full
- Semi
- Anti

Wide → Long

Long → Wide

📊 Visualization Gallery

Chart
Histogram

Live chart

Grouped Summary

□ Hypothesis Tests & Regression

Test Output

Coefficient Table

Scatter + Linear Fit

Diagnostic Plots

ANOVA + Chi-Square

⏳ Time Series Analysis

Forecast horizon (months)

Series + Forecast

Decomposition

ACF / PACF

ADF Test

Forecast Values

K-Means Clustering & PCA

Elbow Method

Cluster Visualisation (k = 3)

Cluster Summary

PCA Scree

PCA Biplot

Variance Table

Reduced Data (First 4 PCs)

🤖 KNN, Logistic Regression & SVM

🎯 KNN — Predict New Customer

Age 35
Income (KES) 85000
Loan Amount (KES) 20000
Credit Score 700
Income > 43,000 → likely **Yes**
Predict

Prediction

KNN — Accuracy vs K

KNN — Confusion Matrix (k = 5)

KNN — Test Predictions

Logistic — Predict Diabetes

Age 45
Hypertension
No
Heart Disease
No
BMI 27
HbA1c 5.5
Blood Glucose 140
Predict

Prediction

Logistic — ROC

Coefficients

Confusion Matrix

SVM — Classify Iris Flower

Sepal Length 5.1
Sepal Width 3.5
Petal Length 1.4
Petal Width 0.2
Predict

Prediction

SVM — Confusion Matrix

Random Forest — Regression & Classification

RF Regression — Predict MPG

Cylinders 4
Displacement 120
Horsepower 120
Drat 3.5
Weight 2.5
1/4 mile time 18
V/S (0/1) 0
Transmission (0/1) 0
Gears 4
Carburettors 2
Predict MPG

Predicted MPG

Feature Importance

Actual vs Predicted (Test Set)

Metrics

RF Classification — Predict Iris Species

Sepal Length 5.1
Sepal Width 3.5
Petal Length 1.4
Petal Width 0.2
Predict Species

Predicted Species

Confusion Matrix (Iris)

⚡ XGBoost — Diabetes Classification

🎯 Predict Diabetes with XGBoost

Age 45
Hypertension
No
Heart Disease
No
Smoking History
never
BMI 27
HbA1c 5.5
Blood Glucose 140
Predict

Prediction

ROC Curve

Feature Importance

Metrics

💡 Explainable AI — Why did the model decide?

🎯 Choose a car to explain

MPG 21
Cylinders 6
Horsepower 110
Weight 2.6
Explain

Prediction

Global Feature Importance

Local Contribution — Why this prediction?

🛒 Project — E-Commerce Transactions

Revenue by Category

Payment Method Frequency

Total Amount Distribution

Monthly Revenue Trend

Spend by Membership

Price vs Total Amount

Revenue Summary